RCS Day 1 - Residential Systems & Shop Safety

CTS Cohort 6 | D09 / RCS01 | Monday, July 20, 2026

Core rule: A professional does not guess. A professional checks, protects, and verifies.

Architecture of the Day

- Theme: Residential construction is a set of connected systems: structure, electrical, plumbing, envelope, and finish.
- Core idea: Students begin RCS by learning how professionals move safely, account for tools, and verify work before trusting it.
- Primary lab: Rolling sink module - assemble, inspect, leak test, troubleshoot, document, and reset.
- Capstone thread: Today introduces the systems thinking and safety behavior needed for each student's individual wall project.
- Room tone: Calm, practical, low-drama, one task at a time. No shame, no rushing, no guessing.

Learning Targets

- Identify the major systems in a residential building.
- Select appropriate PPE for common shop and lab tasks.
- Explain stop-work authority and tool accountability.
- Identify common power tools and their primary hazards.
- Assemble a simple sink drain and P-trap system in a rolling lab module.
- Perform a leak inspection, correct issues, and document verified work.

Materials

- Safety glasses
- Hearing protection
- Gloves for material handling, not rotating tools
- Tape measures
- Pencils/markers
- PPE checklist
- Tool checkout sheet
- Drill/driver, impact driver, circular saw, miter saw, jigsaw, oscillating tool for demonstration
- Rolling sink lab modules
- Faucets, sink basins, tailpieces, P-traps, slip nuts, washers, drain pipe sections
- Buckets/towels
- Student packets
- Verification logs

- Exit tickets

Daily Schedule

Time	Block	Instructor Action
10:00-10:10	Arrival + setup	Students get notebooks, pencils, quiz sheet. Instructor sets out PPE and rolling sink modules but keeps water off.
10:10-10:20	Safety reset + day goal	Frame RCS as residential systems plus professional habits. Tool use is earned by demonstrated control.
10:20-10:35	Morning quiz	Individual, quiet. Use as a teaching check, not a trap.
10:35-10:50	Quiz review	Correct together. Ask students to explain why a safety choice matters.
10:50-11:15	Slides part 1	RCS overview, building systems, final wall capstone preview.
11:15-11:45	Safety + PPE demo	PPE selection, stop-work authority, tool accountability.
11:45-12:05	Power tool intro	Drill/driver, circular saw, miter saw, jigsaw, oscillating tool. Demonstration only unless students are cleared.
12:05-12:30	Whiteboard challenge	Map a simple sink system: supply -> shutoff -> faucet -> sink -> drain -> P-trap -> outlet.
12:30-1:00	Lunch / reset	No unsupervised tool or water work during lunch.
1:00-1:15	Lab demo	Show parts, washer orientation, hand-tight assembly, leak-test setup, and documentation expectation.
1:15-2:15	Rolling sink lab	Teams inventory, assemble, inspect, leak test, correct, and document.
2:15-2:35	Troubleshooting + retest	Groups diagnose leaks or alignment problems, then retest.
2:35-2:50	Cleanup + evidence	Water off, station dry, parts counted, tools returned, verification log completed.
2:50-3:00	Exit ticket + dismissal	Students answer four prompts and preview Day 2 measurement/framing.

Opening Script

- Welcome students by name when possible. Keep the first five minutes low pressure.
- Say: "Today starts Residential Construction Specialty. We are moving from solar systems into building systems, but the rule stays the same: we verify before we trust."
- Say: "Tools do not care how confident you feel. They care where your hands, eyes, material, and attention are."
- Say: "The quiz tells me what I need to reteach. It is not a trap."

Power Tool Introduction Boundary

- Today introduces common shop power tools. It does not automatically clear students for independent use.
- A student earns tool use through demonstration: PPE, inspection, setup, body position, controlled operation, shutdown, and cleanup.

- No student operates a saw independently on Day 1 unless the instructor has directly cleared that student and the task.

Required Stop Conditions

- Any student is unsure of the next step.
- A tool is damaged, missing a guard, or behaving oddly.
- PPE is missing or inappropriate for the task.
- A cut path, bit path, cord path, or water path is unclear.
- Hands, clothing, jewelry, or another person are in the danger zone.
- Water leaks onto a walking surface or near electrical equipment.
- The group cannot explain the system path before testing.

Afternoon Lab: Rolling Sink Module

1. Assign roles: Builder, Inspector, Documenter. Rotate if time allows.
2. Inventory parts and identify each component's purpose.
3. Dry-fit the assembly before tightening.
4. Assemble by hand first; avoid overtightening plastic fittings.
5. Stop for inspection before water testing.
6. Leak test slowly with bucket/towels ready.
7. Diagnose any leak: alignment, washer, slip nut, missing part, or overtightening.
8. Correct one issue at a time, then retest.
9. Document verified work and unresolved questions.
10. Disassemble/reset station only after instructor approval.

Common Mistakes to Watch For

- Cross-threaded slip nut
- Missing or reversed washer
- Trap arm out of alignment
- Overtightening plastic fittings
- No towel/bucket before testing
- Repair made without documenting the leak first
- Students crowding the module instead of using roles

End-of-Day Closeout

- All water off.
- Modules dry.
- Tools and parts counted.
- Floor clear of slip/trip hazards.
- Student verification logs collected or checked.
- Exit tickets collected.

- Instructor notes captured for Day 2 adjustments.

RCS01 Whiteboard Plan

CTS Cohort 6 | D09 / RCS01 | Monday, July 20, 2026

Core rule: A professional does not guess. A professional checks, protects, and verifies.

Board Layout

Board Zone	Content
Left	Today's Rule: Check - Protect - Verify
Center	Residential Systems Map: Structure | Electrical | Plumbing | HVAC | Envelope | Finish
Right	Sink Path: Source -> Shutoff -> Faucet -> Sink -> Tailpiece -> P-trap -> Drain outlet
Bottom	Stop Conditions: unsure, damaged tool, missing PPE, unclear path, leak, cannot explain next step

Draw This Diagram

Water source -> Shutoff -> Faucet -> Sink basin -> Tailpiece -> P-trap -> Drain outlet

Label: supply side, fixture, drain side, trap seal, service point.

Instructor Prompts

- What does each part do?
- Where could this system leak?
- What should be inspected before testing?
- What would make the next person trust your work?

RCS01 Google Classroom Post

CTS Cohort 6 | D09 / RCS01 | Monday, July 20, 2026

Core rule: A professional does not guess. A professional checks, protects, and verifies.

Post Title

RCS Day 1 - Residential Systems, Shop Safety, and Rolling Sink Lab

Post Body

Today we started Residential Construction Specialty. We reviewed how residential buildings work as connected systems: structure, electrical, plumbing, HVAC, envelope, and finish materials.

Main idea: a professional does not guess. A professional checks, protects, and verifies.

Hands-on work: students assembled a rolling sink module, inspected the drain and P-trap path, leak-tested the assembly, corrected problems, documented what was verified, and reset the station.

Exit ticket: name one building system, one safety rule, one verified lab item, and one skill you want more practice with.

RCS01 Quiz Answer Key

CTS Cohort 6 | D09 / RCS01 | Monday, July 20, 2026

Core rule: A professional does not guess. A professional checks, protects, and verifies.

Question	Answer / Rationale
1	B
2	A
3	True
4	Accept: structure, electrical, plumbing, HVAC, envelope, finish.
5	B
6	B
7	B
8	D
9	Accept: checking tools out/in, inspecting condition, returning them clean, reporting missing/damaged tools.
10	checks, protects, and verifies. Accept close phrasing.

RCS01 Master Record

CTS Cohort 6 | D09 / RCS01 | Monday, July 20, 2026

Core rule: A professional does not guess. A professional checks, protects, and verifies.

Package Contents

- Slides/RCS01_Slides.pptx
- Instructor/RCS01_RunSheet.docx
- Instructor/RCS01_WhiteboardPlan.docx
- Instructor/RCS01_AnswerKey.docx
- Instructor/RCS01_ClassroomPost.docx
- Student/RCS01_StudentGuide.docx
- Student/RCS01_Quiz.docx
- Student/RCS01_Lab.docx
- Student/RCS01_VerificationLog.docx
- Student/RCS01_ExitTicket.docx

Source Basis

- Current Cohort 6 operating plan: RCS begins on CTS Day 9 and continues the established Cohort 6 visual/instructional pattern.
- Existing RCS curriculum: 8-day residential construction specialty sequence, including safety, framing, electrical, plumbing, drywall, integrated build, and final assessment.
- Current instructor update: capstone is individual 4 ft x 3 ft wall section with window opening, two receptacles, one switch/light circuit, and drywall cutouts.
- Day 1 lab: rolling sink module with P-trap assembly and leak testing.

Assumptions

- Class time is approximately 10:00 AM to 3:00 PM.
- Students are not independently cleared on saws on Day 1 unless instructor has directly demonstrated and approved use.
- Rolling sink modules and plumbing parts are available or can be staged before lunch.
- Electrical work in RCS Day 1 is only mentioned as a building system, not performed as live wiring.

Revision Notes

- Created as Cohort 6 RCS Day 1 production template.
- Uses short Windows-safe naming convention.
- Integrates professional habits, power-tool safety, stop-work authority, and verification culture.

RCS01 Student Guide

CTS Cohort 6 | D09 / RCS01 | Monday, July 20, 2026

Core rule: A professional does not guess. A professional checks, protects, and verifies.

Big Idea

Residential construction is not just hammering boards together. A building is a set of connected systems. Good tradespeople understand the system, work safely, verify what they did, and leave the work ready for the next person.

Major Residential Systems

System	Plain Meaning	Example
Structure	What holds the building up	Studs, plates, headers, sheathing
Electrical	How power is routed and controlled	Receptacles, switches, lights, breakers
Plumbing	How water enters and waste leaves	Supply lines, drains, traps, shutoffs
HVAC	How air and heat move	Vents, ducting, filters, heat pumps
Envelope	How the building separates inside from outside	Windows, insulation, air sealing
Finish	What protects and completes surfaces	Drywall, trim, paint, flooring

Professional Safety Habits

- Wear PPE that matches the task.
- Ask before using unfamiliar tools.
- Check the work area before starting.
- Use stop-work authority when unsure.
- Tell the truth about mistakes early.
- Clean as you go.

Power Tool Introduction

- Drill/driver: holes, screws, and fasteners. Main hazards: bit slips, stripped fasteners, pinched fingers.
- Circular saw: straight cuts in sheet goods and lumber. Main hazards: blade path, kickback, support, cord path.
- Miter saw: controlled crosscuts. Main hazards: hand placement, blade coast-down, material support.
- Jigsaw/oscillating tool: cutouts and detail cuts. Main hazards: wandering blade, hidden material, hand placement.

Rolling Sink Lab Purpose

Today's lab teaches systems thinking. You will assemble a simple plumbing drain setup, test it, find leaks if they happen, correct them, and document what you verified.

Today's Field Note

A professional is not the person who never makes a mistake. A professional is the person who notices early, stops safely, fixes cleanly, and leaves evidence that the work was checked.

RCS01 Morning Quiz

CTS Cohort 6 | D09 / RCS01 | Monday, July 20, 2026

Core rule: A professional does not guess. A professional checks, protects, and verifies.

Name: _____ Date: 7/20/2026

1. What does RCS focus on in this module?

- A. Only plumbing
- B. Residential construction systems and professional shop habits
- C. Only drywall finishing
- D. Guessing faster

Answer: _____

2. Which item is usually required when using cutting, drilling, or fastening tools?

- A. Safety glasses
- B. Sunglasses only
- C. Loose gloves around rotating tools
- D. No PPE if the task is quick

Answer: _____

3. True or false: Stop-work authority means students can pause a task when something is unsafe or unclear.

Answer: _____

4. Name two major systems found in a residential building.

Answer: _____

5. What should happen before a student uses an unfamiliar power tool?

- A. Watch one internet video
- B. Ask for instructor demonstration and clearance
- C. Try it quickly
- D. Let the fastest student go first

Answer: _____

6. What is the purpose of a P-trap?

- A. It stores clean water for drinking

- B. It holds water to block sewer gas and creates a service point
- C. It increases water pressure
- D. It replaces the faucet

Answer: _____

7. Why should parts be dry-fit before final tightening?

- A. To make the lab take longer
- B. To check alignment and avoid forcing parts
- C. To skip inspection
- D. To hide leaks

Answer: _____

8. Which is a stop condition during the sink lab?

- A. A small leak appears
- B. A washer is missing
- C. The group cannot explain the path
- D. All of the above

Answer: _____

9. What is tool accountability?

Answer: _____

10. Complete the sentence: A professional does not guess. A professional _____.

Answer: _____

RCS01 Afternoon Lab - Rolling Sink Module

CTS Cohort 6 | D09 / RCS01 | Monday, July 20, 2026

Core rule: A professional does not guess. A professional checks, protects, and verifies.

Lab Mission

Build a simple sink drain and P-trap assembly in the rolling lab module, inspect it, leak-test it, troubleshoot problems, and document verified work.

Safety Boundary

- Water stays off until the assembly has been inspected.
- No electrical tools or cords near wet areas.
- Wipe spills immediately.
- Stop if a fitting does not align or a part is missing.
- Do not overtighten plastic fittings. Hand-tight plus instructor guidance only.

Roles

Role	Job
Builder	Assembles parts and calls out each step.
Inspector	Checks alignment, washers, tightness, and leaks.
Documenter	Records parts, issues, fixes, and evidence photos.

Parts Inventory

Part	Have it?	What it does	Notes
Sink basin/module		Holds water and routes it to drain	
Faucet		Controls water into sink	
Supply line/shutoff		Brings water to faucet and allows shutoff	
Tailpiece		Connects sink drain to trap	
P-trap		Holds water seal and service point	
Slip nuts/washers		Seal adjustable drain joints	
Drain outlet		Carries water away	

Build Sequence

1. Inventory all parts.

2. Lay parts out in system order.
3. Dry-fit the tailpiece, trap, and outlet path.
4. Check washer direction and slip nut alignment.
5. Hand-tighten fittings.
6. Stop for inspection before water testing.
7. Place bucket/towel before testing.
8. Run a small amount of water and observe.
9. Mark or record any leak before fixing it.
10. Correct one issue at a time.
11. Retest and verify dry joints.
12. Reset the module after instructor approval.

Leak Troubleshooting Table

Problem observed	Likely cause	Fix attempted	Retest result	Instructor initials

Success Criteria

- Parts identified and assembled in correct order.
- No leaks after correction and retest.
- Group can explain the water path.
- Station is clean, dry, and reset.
- Verification log is complete.

RCS01 Verification Log

CTS Cohort 6 | D09 / RCS01 | Monday, July 20, 2026

Core rule: A professional does not guess. A professional checks, protects, and verifies.

Student: _____ Group: _____ Date: 7/20/2026

Task / Station	What I expected	What I observed	Method used	What I verified	What still needs checking	Signoff
PPE check						
Tool / part inventory						
Sink assembly inspection						
Leak test						
Cleanup/reset						

Sample Entry

Task: Sink assembly inspection | Expected: washers seated and slip nuts straight | Observed: one washer missing | Method: visual inspection before water test | Verified: corrected washer installed before testing | Still needs checking: leak test after water is on

RCS01 Exit Ticket

CTS Cohort 6 | D09 / RCS01 | Monday, July 20, 2026

Core rule: A professional does not guess. A professional checks, protects, and verifies.

Name: _____ Date: 7/20/2026

One residential system I can name is:

_____

_____

One safety rule I need to remember is:

_____

_____

One thing I verified during lab was:

_____

_____

One thing I want more practice with is:

_____

_____

RCS Day 1

Residential Systems & Shop Safety

CTS Cohort 6 | Monday, July 20, 2026

Today's rule:
A professional does not guess. A professional checks, protects, and verifies.



One building. Many systems. One standard: verify before covering.

Today's flow

Start safe, learn the system, then build and test a simple plumbing assembly.

1	10:00	**Arrival + safety reset**
2	10:15	**RCS overview**
3	10:40	**PPE + shop rules**
4	11:10	**Power tool intro**
5	11:35	**Quiz + review**
6	12:30	**Rolling sink lab**
7	2:35	**Document + cleanup**
8	2:50	**Exit ticket**

Quiet quiz first. Hands-on work after the safety frame is clear.

Learning targets

By dismissal, students should be able to do these without guessing.

Name the major residential building systems.
Select PPE for common shop tasks.
Explain stop-work authority and tool accountability.
Identify common power tools and their main hazards.
Assemble a simple sink drain and P-trap.
Leak test, correct, and document the assembly.

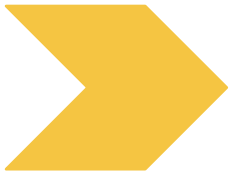
Today is about professional habits before capstone work starts.

What RCS means in CTS

Residential Construction Specialty is where technical habits become jobsite habits.

Work habits

Measure before cutting
Inspect before covering
Communicate before moving material
Clean before calling a task finished



Trade content

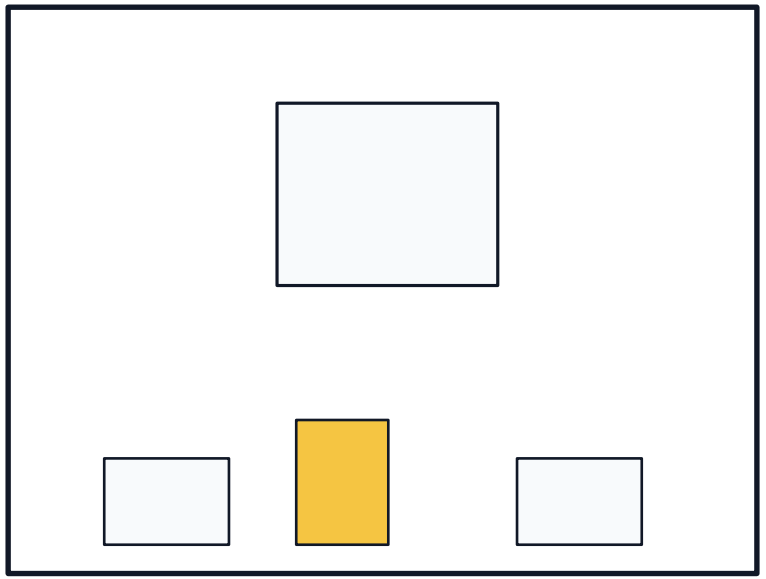
Framing
Electrical rough-in
Plumbing basics
Drywall and inspection

The habit is the portable skill.

The final wall project starts today

Every day points toward the individual 4 ft x 3 ft wall build.

Frame a small wall section.
Build a window rough opening.
Install two receptacle boxes.
Install one switch and light circuit.
Drywall the whole board with proper cutouts.
Explain and inspect the finished work.



4 ft x 3 ft individual wall

Residential systems map

Buildings are not random parts. They are connected systems.

Structure

Electrical

Plumbing

HVAC

Envelope

Finish

A good builder understands what will be hidden, protected, or affected by the next trade.

Professional expectations

Employers notice habits before they trust skill.

Show up ready to work.
Ask before touching unknown tools or systems.
Keep your area organized enough to be safe.
Tell the truth about mistakes quickly.
Document work so another person can understand it.

Today is about professional habits before capstone work starts.

Safety is not a speech

Safety is the way the work gets done.

PPE

Eyes, ears, hands, lungs, and clothing match the task.

Position

Stable feet, clear hands, clear blade path, no body parts in the line of fire.

Communication

Call out before cuts, moving material, energizing, or testing.

PPE for today

Use the protection that matches the hazard. More is not always better; wrong PPE is still wrong.

Eye protection

Required during cutting, drilling, fastening, and leak-test cleanup.

Gloves

Useful for carrying material. Not for rotating blades/bits.

Hearing protection

Required around saws and loud tools.

Dust control

Drywall and cutting dust need cleanup and breathing awareness.

Stop-work authority

Stopping is a skill. Guessing is not a method.

Something feels unsafe or unclear.
A tool is damaged or behaving oddly.
A cut, route, or connection does not match the plan.
Someone is in the path of material, blade, cord, water, or tool movement.
You cannot explain the next step.

Stop. Ask. Verify.

Tool accountability

A missing tool is not a mystery. It is a future problem.

1. **Check out**
2. **Inspect**
3. **Use correctly**
4. **Return clean**
5. **Report damage**

Same tool. Same place. Every time.

Power tool safety ladder

Students earn tool use through demonstrated control, not confidence.

1. **Identify tool**
2. **Inspect condition**
3. **Set material support**
4. **Check body position**
5. **Operate slowly**
6. **Shut down safely**
7. **Clean and return**

Confidence is not clearance. Demonstration is clearance.

Drill and driver safety

Small tools can still injure people and ruin work.

- Correct bit or driver tip
- Battery removed during bit changes
- Clamp or secure material
- Two hands when needed
- Low speed for starting
- Stop before stripping or splitting

Slow is smooth. Smooth is safe.

Saw safety introduction

A saw cut is a planned movement, not a surprise event.

Know the blade path
Support both sides of material
Keep hands outside the danger zone
Wait for full stop before moving off cut
Never bypass guards
Ask before any unfamiliar cut

Slow is smooth. Smooth is safe.

Measurement habit

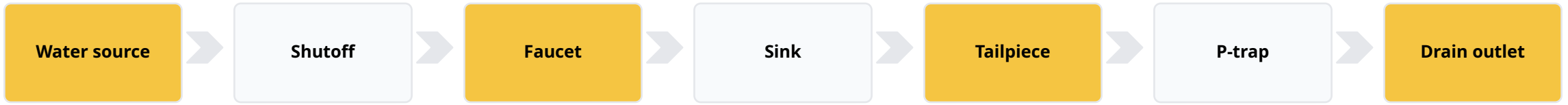
The cut is only as good as the layout.

Read the tape correctly.
Mark the waste side.
Use the square for straight lines.
Verify with a partner before cutting.
Write important dimensions down.

Mark it so someone else can understand it.

Plumbing system example

Today's lab uses plumbing because leaks tell the truth quickly.



Supply brings clean water in. Drainage carries used water out. The trap protects the space.

Why a P-trap exists

The bend is not decorative. It is a simple safety system.

Holds water to block sewer gas.
Creates a removable service point.
Needs correct washer orientation and tight joints.
Leaks usually come from alignment, missing washers, or overtightening.

The part has a job. Know the job before tightening.

Afternoon lab mission

Rolling sink module: assemble, test, correct, document.

1. **Inventory parts**
2. **Dry-fit layout**
3. **Assemble by hand**
4. **Leak test**
5. **Troubleshoot**
6. **Document**
7. **Reset station**

Build - test - correct - document.

Lab roles

Clear roles prevent six hands from becoming one plumbing octopus.

Builder

Assembles parts and calls out each step.

Inspector

Checks alignment, washers, tightness, and leaks.

Documenter

Records parts, problems, fixes, and evidence photos.

Common mistakes

Mistakes are data when they are noticed early.

- Cross-threaded slip nut
- Washer missing or reversed
- Trap arm not aligned
- Overtightened plastic fitting
- No bucket or towel during testing
- Leak not documented before repair

Find it early. Fix it clean.

Verification checkpoint

Stop before water. Inspect before testing.

All washers present and facing correctly.
Slip nuts started straight.
Trap path can be explained.
Bucket/towel ready.
Instructor or peer check complete.

No water until the path makes sense.

Verification log

Documentation is safety support, not paperwork theater.

What I checked
What I expected
What I observed
Tool or method used
What I corrected
What still needs checking

If it is not documented, the next person has to guess.

Cleanup standard

The job is not done until the next person can start safely.

Water off and area dry
Parts counted and returned
Tools cleaned and stored
Floor clear of trip/slip hazards
Evidence uploaded or turned in

Reset is part of the work.

Exit ticket

Write complete sentences. Short is fine. Vague is not.

One residential system I can name is…
One safety rule I need to remember is…
One thing I verified during lab was…
One thing I want more practice with is…

This tells us what to reteach.

Tomorrow preview

Day 2: measuring, layout, framing, and controlled tool use.

Tape measure accuracy
Speed square layout
Cut list basics
Wall parts: plates, studs, rough opening
Power tool clearance and supervised cuts
Building toward the individual wall project

Tomorrow starts the wall.